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Information hiding in edges: A high capacity information hiding technique using hybrid edge detection

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Abstract The multimedia security is becoming more and more important as the data being exchanged on the Internet is increasing exponentially. Though cryptography is one of the methods which is used to secure the data during transit, but the camouflaged appearance of the scrambled data alerts the adversary about some critical information being shared. In such a scenario, steganography has been used as an alternate solution to secure the secret information. In this paper a color image steganographic algorithm based on hybrid edge detection is proposed. The color image is partitioned into constituent Red (R), Green (G) and Blue (B) planes. Hybrid edge detection is used for finding the edge and non-edge pixels of Green and Blue planes of cover image. The Green and Blue planes are used for hiding the data while Red plane holds the pixel status (whether edge or non-edge) of these planes. The RC4 encryption algorithm is used to encrypt secret message before embedding it in the cover image to enhance security of the secret data. A fragile watermark/logo (whose size is less than 1% of total secret data) has been embedded, besides secret data in the cover image, to facilitate content authentication and early tamper detection. At the receiver, firstly logo is extracted. If it is same as one embedded at transmitter, indicating that secret data has not been altered during transit, secret data is extracted. Otherwise (if extracted logo is not same as used at input) the receiver does not waste critical time to extract compromised data but sends an automatic retransmission request. Experimental investigations reveal that the proposed scheme is capable of providing high quality of stego-images for a fairly high pay load. A comparison of the proposed technique with some state of art schemes substantiates the above arguments.

Keywords Canny edge detection · Prewitt edge detection · Pixel value differencing (PVD) · Hybrid edge detection · Human visual system (HVS) · Ciphering

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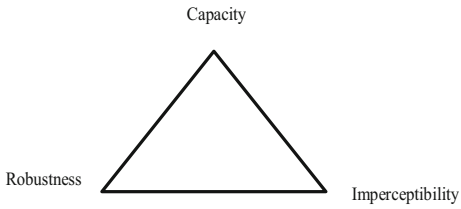
1 Introduction

The recent advancement in the technology and multimedia network infrastructure has resulted in the piracy of the information being sent over the network infrastructure. Copying information and using it illegally has become very easy due to current state of art technology. Thus the given scenario poses a series of challenges for the research community so as to find ways to ensure security of sensitive secret data. In such a scenario various techniques like cryptography and steganography are being used to save secret data from the adversary. Cryptography scrambles the data and makes it obscure and as such adversary can easily get to know that some spatial information is being transmitted [12]. Steganography hides the existence of data and has flourished as an important tool to secure secret data from adversary [3]. However if cryptography and steganography are combined it will provide the best cloistered communication.

Steganography is the science of hiding information into innocuous objects in such a manner that the existence of the secret information remains imperceptible to an external adversary. The innocuous object itself is called the cover and the hidden information in it the payload. The choice of cover ranges from simple text files to images, audio and video [16]. Images are widely used as a cover for steganography because of its pervasiveness in daily applications and high redundancy in representation. Steganography is method of invisible communication between two parties. Steganography is more concerned with payload rather than robustness against intentional attacks unlike watermarking. The most important evaluation criteria's for any good steganography techniques are Capacity, Imperceptibility, Security and Robustness [10]. Unfortunately, these main requirements contradict with each other as depicted in famous conflict triangle shown in Fig. 1. Capacity is the number of secret bits embedded in cover image without deteriorating its perceptual quality. High capacity steganographic system is one in which more secret bits can be hidden in the cover image without deteriorating its perceptual quality. Imperceptibility is usually measured in terms of Peak Signal-to-Noise Ratio (PSNR). If the difference between the cover image and the stego-image is small it means that PSNR value is high. For a given scenario, trade-off between the capacity, imperceptibility and robustness are worked out in accordance with the Fig. 1.

Image steganographic techniques are classified into spatial and transform domain techniques. Spatial domain methods apply direct pixel manipulation for information hiding. They are also characterized by simplicity, shortened implementation time, reduced hardware requirement and overall low time complexity. However, in frequency domain method cover image is first transformed in frequency domain and then the secret message is embedded into the transformed cover image. The fundamental and most popular method for message concealment in digital images is Least Significant Bit (LSB) insertion. This scheme because of ease of implementation and high payload capability is popular, however can easily be broken by adversary. Data hiding in edge areas of an image is an interesting proposition. Lot of work is being carried out in this area. Some of the state of art work has been discussed in the following section. Rest of paper is organized as follows: Section 2 presents a

Fig. 1 Conflict triangle



comprehensive literature survey of relevant work. The proposed scheme has been discussed in section 3. Section 4 presents experimental results and discussions. The paper concludes in section 5.

2 Related work

Owing to fast changing trends in the field of multimedia technology, information security is one of the most important areas of research. Data hiding, due to its inherent capability to avert adversaries is getting attention of researchers across the globe, for securing information. The most widely used and the simplest method for hiding the secret information in cover image (in image steganography) is Least Significant Bit (LSB) technique [15]. In this method LSB of pixel of cover image is changed according to secret message. However if size of secret message is large then two or more bits of the pixels of cover image has to be changed which deteriorates the quality of stego-image. With an aim to improve payload, security and imperceptibility many variants of LSB techniques have been reported so far [5, 13, 21]. However, these techniques do not resist any statistical attacks [1]. Many spatial domain techniques for high capacity and better imperceptibility besides ability to with stand LSB removal/replacement attack have been reported in [18]. Wu DC and Tsai WH [22] proposed an image steganography method known as Pixel Value Differencing (PVD). This method is based on difference value between the two consecutive pixels. In this method cover image is partitioned into two non-overlapping consecutive pixel blocks. The difference value is obtained from the values of the two pixels in each block. This difference value is modified according to the secret message. This method provides better results of PSNR and embedding capacity. However this method is susceptible to histogram based attacks. Chung MW et al. [8] put forward a method for data hiding which is based on pixel value differencing and modulus function. Here difference value between the consecutive pixels of cover image is calculated then the modulus function is performed on that difference value. The remainder of modulus function is modified according to secret message instead of modifying difference value of consecutive pixels. This approach reduces the distortion caused due to hiding the secret data in cover image. Chin-Chen Chang and Hsien-Wen Tseng [7] proposed a method in which difference value is calculated between Upper and Left neighboring pixels which provides better compression. This difference value is used to embed the secret data in cover image in raster-scan order except first row and first column of the cover image. In this scheme, if difference is -1, 0, 1, then only one bit of secret message is embedded in the cover image and for other values of difference more than one bit of secret message is embedded.

Human Visual System (HVS) is more sensitive to slight changes in smooth areas than edge regions [20]. That is, an edge area can hide more secret data than a smooth area [4]. So, it is difficult to find visual artifacts and detectable artifacts in the edge areas after secret data hiding. Saurabh & Ashutosh [14] have proposed steganography method for color image using the Canny edge detection and Hashing. In this method the cover image is partitioned into two pixel blocks of edges and non-edges. Then the hash function decides where secret message is to be embedded. The Hash function uses secret message, cover image and Hash Key (K) which is generated by using the edge pixels of the cover image to produce sequence of hash values and these hash values determine pixels in the cover image where secret message is to be embedded. The PSNR of stego-images obtained using this method is 47.55 dB. Jae G. Y. et al. [11] have proposed a high capacity data hiding technique based on edge detection and 2^k correction. The authors have reported an average PSNR value of about 42 dB for a payload of 398,720 bits. Anastasia et al. [2] have proposed a data hiding algorithm which is based on Random Number Generator (RNG) for embedding the data. In this method Hybrid edge detector is used to determine edge and non-edge pixels of Red (R), Green (G) and Blue (B) planes of cover

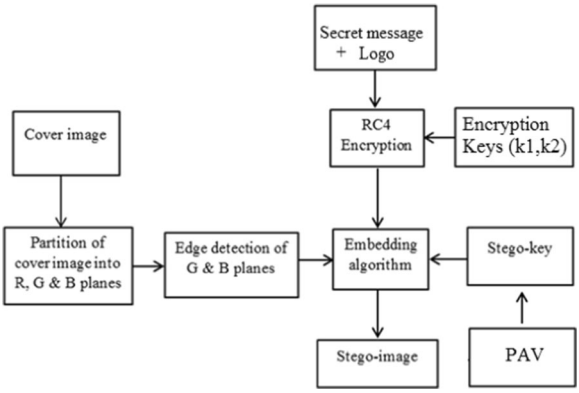
image. The RNG decides the number of bits of secret data to be embedded in pixels of R, G and B planes of cover image based upon their nature (edges or non-edges). The scheme is reported to hide a fair amount of data into cover images without deteriorating their quality significantly.

A hybrid edge detector based data hiding scheme has been proposed in [6]. Canny edge detector and Fuzzy edge detector has been used to detect edge and non-edge pixels. The image is then partitioned into n- pixel blocks i. e. [$P_1, P_2, P_3 \ldots\ldots P_n$]. The first pixel block P1 is used to hold status of remaining pixels in the block whether they are non-edge or edges. The status reflects '1' if P_i is the edge pixel however it reflects '0' if P_i is the non-edge pixel. The LSBs of P_1 are used to hold status pixels from P_2 to P_n. The n-pixel block is separated into two classes one class belongs to non-edge pixels and another class belongs to edge pixels. In the non-edge pixel block, 'x' bits of secret information are embedded in each pixel and in edge pixel block, 'y' bits of secret information are embedded in each pixel. The values n are chosen to be 3, 4 and 5. The values of 'x' are chosen 1 or 2 in most of the cases and the values of 'y' are 2, 3, 4 or 5. This method doesn't show good results of PSNR and image imperceptibility when the value of n is increased beyond 4. Though the above mentioned scheme provides good quality of stego-images but as the block size increases the perceptual quality of the stego-images is worsened. With an aim to improve imperceptibility of stego- images for a fair payload a high capacity data hiding technique based on hybrid edge detection has been presented in this paper. RC4 stream cipher with a key size of 128 bits has been used for encrypting the secret data before embedding to enhance security of the given system.

3 Proposed system

The proposed data hiding method is shown in Fig. 2. The cover image is partitioned into Red (R), Green (G) and Blue (B) planes. Edge detection is used to classify the cover image pixels into edge and non-edge ones. Hybrid edge detector consisting of Canny and Prewitt detector has been used to maximize the number of detected edges and hence the payload. Section 3.1 presents a detailed discussion on Prewitt and Canny edge detection. Since a color image pixel composes of three constituent pixels (R, G and B); G and B pixels of every color pixel of a test image are used for secret data embedding while as R plane is used to hold the status of G and B plane pixels (whether edge or non-edge). In order to enhance the security of embedded secret data, it is encrypted using RC4 algorithm [9]. RC4 algorithm because of simplicity and efficient encryption provides more security to the system. It is pertinent to mention that two encryption keys (K1, K2) have been used here to encrypt logo and secret data respectively. This has been done to ensure that logo is successfully extracted independent of secret data if and when required.

Fig. 2 Block diagram



For detailed description of RC4 algorithm please refer to [19]. The secret data and logo has been embedded under the control of stego- key generated by Pseudorandom Address Vector (PAV) as described in [17]. The procedure of embedding the secret message in the cover image and extracting the secret message from stego-image is discussed in sections 3.2 and 3.3 respectively.

3.1 Edge detection

The proposed data hiding scheme uses edge detection for locating the edges of cover image. An edge indicates the boundary between the actual object and background. Edge detection process holds the most significant place in the proposed scheme. There are many edge detection techniques like Sobel, Prewitt, Canny, Fuzzy and Laplacian [4]. Hybrid edge detector consisting of Prewitt and Canny edge detection techniques has been used in this work. The Prewitt edge technique is used to detect the edges of images by convolving it with a small, separable and inter-valued filter in horizontal and vertical directions which makes it relatively inexpensive from computational complexity point of view. Canny edge detector is the one of the mostly trusted method that provides reliable edge detection. The Prewitt edge and Canny techniques are discussed briefly as under:

3.1.1 Prewitt edge detection

The Prewitt edge method is performed in following steps:

1. In the first step two 3 × 3 kernel are convolved with the cover image to calculate derivate for horizontal changes and vertical changes. If A is the cover image, G_x and G_y are horizontally and vertically decomposed images at each point. The G_x and G_y are computed as

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} *A \text{ and } G_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} *A$$

   Where * denotes the 2D convolution operation

2. The kernels obtained in vertical and horizontal direction are used to calculate absolute magnitude of the gradient at each point using Eq. (1) :

$$|G| = \sqrt{Gx^2 + Gy^2} \tag{1}$$

   However an approximate magnitude is computed by using:

$$|G| = |G_x| + |G_y| \tag{2}$$

   The angle of orientation (θ) of edge is given by Eq. (3)

$$\theta = \arctan\left(\frac{Gy}{Gx}\right) \tag{3}$$

Figure 3 shows results of ‘G’ and ‘B’ planes of various test images obtained by Prewitt edge detection process. It is pertinent to mention that PE denotes number of edge pixels in an image.

3.1.2 Canny edge detection

The canny edge process is performed in following steps:

1. In order to remove false edges, the image is first smoothened by using Gaussian filtering. The kernel of Gaussian filtering with standard deviation $\sigma = 1.4$ is shown in Eq. (4)

$$B = \frac{1}{159}\begin{bmatrix} 2 & 4 & 5 & 4 & 2 \\ 4 & 9 & 12 & 9 & 4 \\ 5 & 12 & 15 & 12 & 5 \\ 4 & 9 & 12 & 9 & 4 \\ 2 & 4 & 5 & 4 & 2 \end{bmatrix} \tag{4}$$

2. The next step after smoothening is to find the edges by finding the gradients of the image. Sobel-operator is used to determine the edges of smoothened image. The kernels given in Eqs. (5 and 6) are used to approximate the gradient in X- and Y direction

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \tag{5}$$

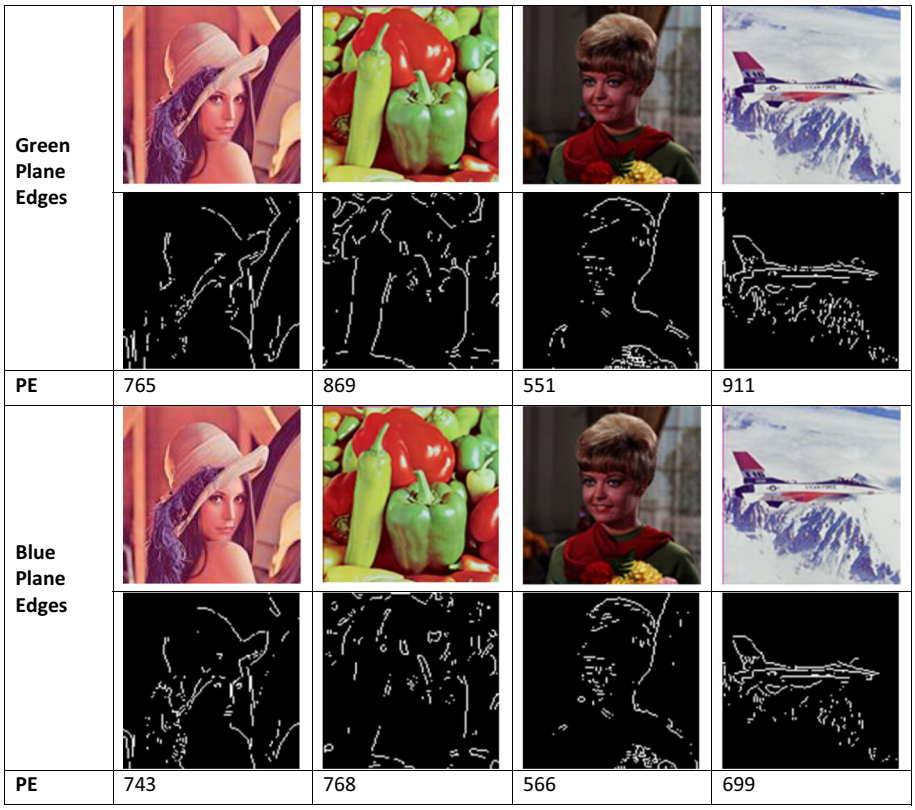


Fig. 3 Edges detected by Prewitt edge detector for Green and Blue planes

$$G_y = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \tag{6}$$

The magnitude of the gradient is calculated using Eq. (7)

$$|G| = \sqrt{Gx^2 + Gy^2} \tag{7}$$

3. After calculating the gradient, non-maximum suppression is used to 'thin' the edges.
4. In order to differentiate the true edges and false edges of the image a threshold is applied. The pixels stronger than threshold are marked as edges and those weaker than threshold are marked as non-edges.

The edges detected by Canny edge detector for 'G' and 'B' planes of various test images are depicted in Fig. 4.

It is evident from subjective and objective quality of Figs. 3 and 4 that the performance of the two edge detectors is not same, the performance of Canny edge detector being better. In order to improve the edge detection further we have used hybrid edge detection process. The hybrid edge detector used for this research work consists of Canny and Prewitt edge detectors. Both the detectors return a binary matrix for an input image. The binary '0' and '1' in such detector outputs represent non-edge and edge pixels respectively. Logical OR function (of Canny and Prewitt detector outputs) is used to compute the total edge pixels in a cover image. This process is depicted as shown in Fig. 5.

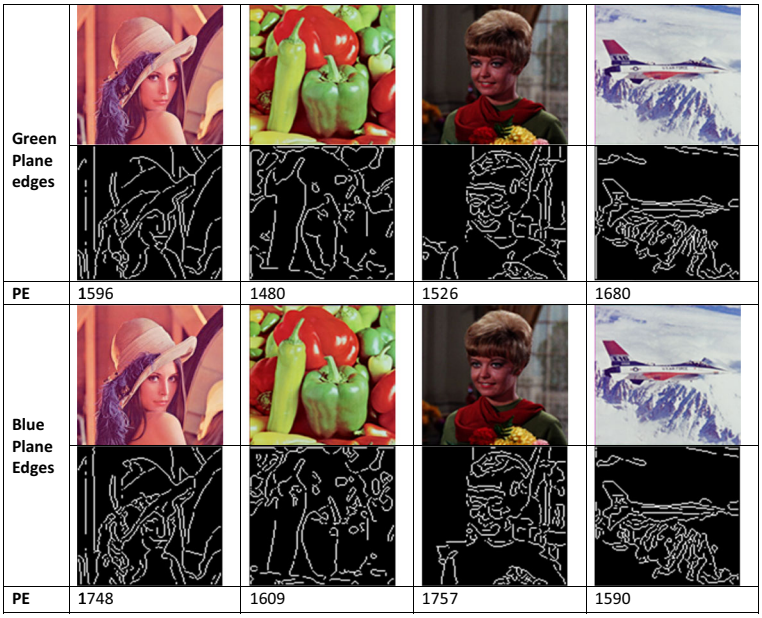


Fig. 4 Edges detected by Canny edge detector for Green and Blue planes

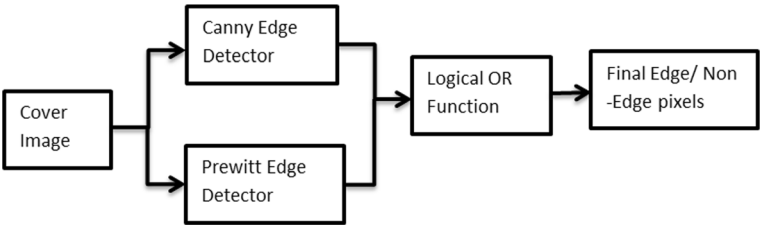


Fig. 5 Hybrid Edge Detector

The performance of the hybrid edge detector could be seen from the Fig. 6. It is clear that this detector is capable of finding more edges than that of Prewitt and Canny when used separately.

3.2 Embedding algorithm

The embedding is carried out as per following procedure:

1. Firstly, the cover image (C) is partitioned into R, G and B planes.
2. G and B planes are passed through hybrid edge detector so as to detect edged and non-edged pixels of G and B. The output of hybrid edge detector for each

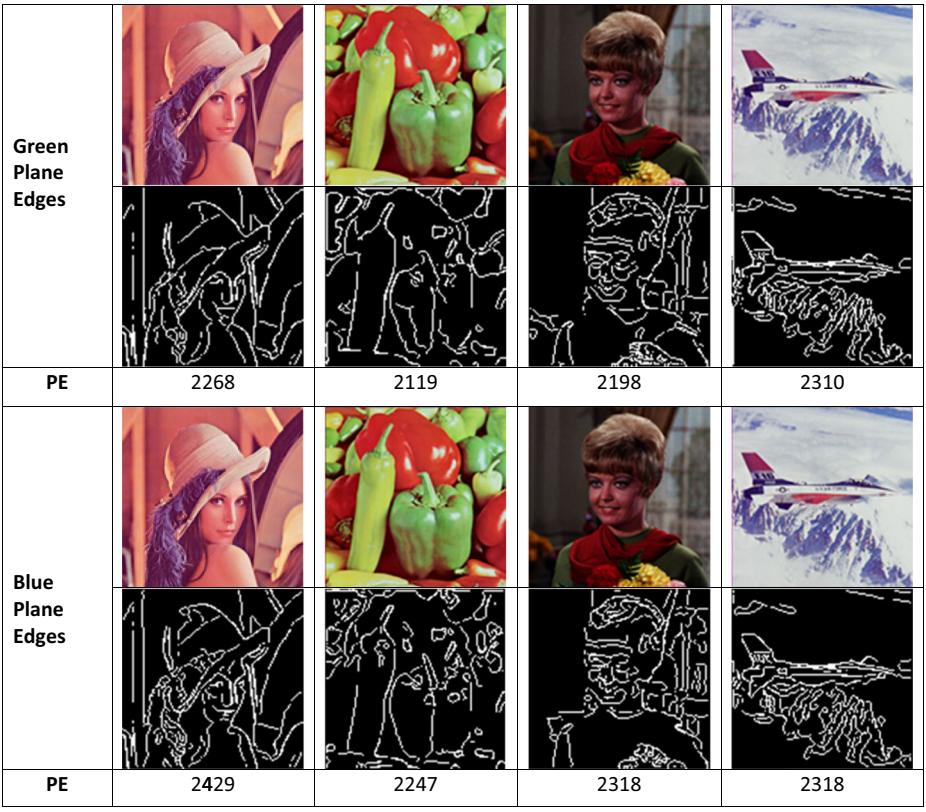


Fig. 6 Edges detected by Hybrid edge detector for Green and Blue planes

of the planes is a binary matrix of same dimensions as that of input image (C). The respective binary matrices are represented by G′ and B′. The Output of edge detector indicates whether a given pixel is edged or non-edged. A '0' at location $B'_{(i,j)}$ represents that the pixel of given G plane located at $G_{(i,j)}$ is a non-edge pixel while '1' represents that it is an edge pixel. Similarly, a '0' at location $G'_{(i,j)}$ represents that the pixel of B plane located at $B_{(i,j)}$ is a non-edge pixel while '1' represents that it is an edge pixel.

3. Estimate the maximum number of message bits that can be hidden in a given image, let it be 'n'. Divide 'n' into two classes: i) logo bits (1024 bits used in our algorithm) ii) Secret data bits 'm'.
4. Divide the message bits in four blocks each of m/4 bits.
5. Encrypt logo bits by encryption key K1.
6. Encrypt each block of message bits using RC4 encryption technique using (K2).
7. For embedding encrypted data in cover image $C_{(i,j)}$, the status of constituent pixels $G_{(i,j)}$ and $B_{(i,j)}$ is checked by checking the values of $G'_{(i,j)}$ and $B'_{(i,j)}$. The status is stored in LSBs of pixel $R_{(i,j)}$. It is pertinent to mention here that LSB of R stores the status of Green plane pixel and first Intermediate Significant Bit (ISB) stores the status of Blue plane pixel. The LSBs of R plane pixels are modified according to the scheme shown in Table 1.
8. Embed logo bits at first 512 (or lesser) locations pointed to by PAV in accordance with Table 1. 512 addresses are required for (1, 1) scheme as in this case two bits would be embedded in one pixel (G and B planes) at a given location. For (1, n) with n > 2 lesser addresses are required.
9. Embed one bit in non-edge pixel and multiple bits (2-6) in edge pixels. The data is embedded at the addresses pointed to Pseudorandom Address Vector which is used as a stego key.
10. The process is repeated until whole secret message is embedded in the cover image.

3.3 Extraction algorithm

1. The stego image is partitioned into R, G and B planes in the first step.
2. The status of LSB and first ISB bits of $R_{stego(i,j)}$ pixels of R plane are scanned and Pseudorandom Address Vector used to extract the embedded secret message from the stego-image. If status of LSB and ISB bits of $R_{stego(i,j)}$ pixel is '00', this means that both pixels are non-edge pixels, so one bit of secret message is extracted from each G and B planes of stego-image otherwise one bit or 1-6 bits of secret message are extracted depending upon the status LSB and first ISB of $R_{stego(i,j)}$ pixel of R plane of cover image (from the addresses pointed to by PAV). The extraction is carried out as follows

Table 1 Red plane LSB modification

Green plane pixel status	Blue plane pixel status	Red plane LSBs (two) modification
Non-edge	Non-edge	00
Non-edge	Edge	01
Edge	Non-edge	10
Edge	Edge	11

i) Firstly extract logo bits using corresponding PAV addresses.
ii) Decrypt logo bits with K1
iii) If extracted logo is same as one embedded go to step 3.

3. The step 2 is repeated until whole secret message (m bits) is extracted from the stego image.

4. Divide the extracted message bits in four blocks each of m/4 bits.

5. RC4 decryption technique is used to decrypt the message blocks with the same key as used during encryption process.

The data embedding and extraction process have been explained using following examples:

4 Example 1: data embedding

Consider an arbitrary 4 × 4 block of test image 'Lena' with its constituent planes represented as:

$$\mathrm{R} = \begin{bmatrix} 11000111 & 11100001 \\ 11100110 & 11100111 \end{bmatrix}, \mathrm{G} = \begin{bmatrix} 111011011 & 11001101 \\ 110011101 & 11001101 \end{bmatrix} \text{and B} = \begin{bmatrix} 11001101 & 00011100 \\ 11100011 & 11100110 \end{bmatrix}.$$

G and B planes are passed through hybrid edge detector. Let us assume that by hybrid edge detector it is found that $G_{(1,1)}$, $G_{(2,2)}$, $B_{(1,2)}$ and $B_{(2,1)}$ are edge pixels and $\mathrm{G}_{(2,1)}$, $\mathrm{G}_{(1,2)}$, $\mathrm{B}_{(1,1)}$ and $\mathrm{B}_{(2,2)}$ are non-edge pixels. The output of hybrid edge detector would be $G' = [1 \quad 0 \;\; 01]$ and $\mathrm{B}' = [0 \quad 1 \;\; 10]$.

The status of the B and G planes is stored by R plane of stego- image and is given by:

$$\mathrm{R}_{\mathrm{stego}} = \begin{bmatrix} 11000\mathbf{01} & 11100\mathbf{10} \\ 11100\mathbf{10} & 11100\mathbf{01} \end{bmatrix}$$

Let the secret message to be embedded be M = [000111010111]. If we embed one bit of secret message in non-edge pixels and two bits in edge pixels; the G and B planes of cover image will be modified as:

$$\mathrm{G}_{\mathrm{stego}} = \begin{bmatrix} 111011\mathbf{00} & 1100111\mathbf{1} \\ 11001110 & 11001\mathbf{11} \end{bmatrix}$$

$$\mathrm{B}_{\mathrm{stego}} = \begin{bmatrix} 11001\mathbf{10} & 00011\mathbf{11} \\ 11100\mathbf{10} & 11100\mathbf{11} \end{bmatrix}$$

5 Example 2: data extraction

For data extraction, consider the stego-image having R, G and B planes as:

$$R_{stego} = \begin{bmatrix} 11000101 & 11100010 \\ 11100110 & 11100101 \end{bmatrix},$$

$$G_{stego} = \begin{bmatrix} 11101100 & 11001111 \\ 11001101 & 11001111 \end{bmatrix}$$

$$B_{stego} = \begin{bmatrix} 11001100 & 00011111 \\ 11100010 & 11100111 \end{bmatrix}.$$

By checking the status of LSB and first ISB of pixels $R_{(11)}$, $R_{(12)}$, $R_{(2,1)}$, *and* $R_{(2,2)}$ of R_{stego} plane of Stego-image, which is 01,10,10 and 01 respectively, it is reflected that $G_{(1,1)}$, $G_{(2,2)}$, $B_{(1,2)}$and $B_{(2,1)}$ are edge pixels and $G_{(2,1)}$, $G_{(1,2)}$, $B_{(1,1)}$ and $B_{(2,2)}$ are non-edge pixels of G_{stego} and B_{stego} planes of the stego-image, So two bits of secret message are extracted from $G_{(1,1)}$, $G_{(2,2)}$, $B_{(1,2)}$and $B_{(2,1)}$ pixel and one bit of secret message is extracted from $G_{(2,1)}$, $G_{(1,2)}$, $B_{(1,1)}$ and $B_{(2,2)}$ pixels. The extracted secret message will be $M_{extracted}$ = [000111010111].

6 Results and discussion

The proposed scheme has been tested on various standard color images of size 128 × 128, 384 × 512 and 512× 512. The scheme has been implemented on i3-2350 processor with 2GB RAM on MATLAB 2014b platform. The quality of the stego-image has been tested using various indices like PSNR, Normalized Cross-Correlation (NCC), Mean Square Error (MSE)

Fig. 7 Various 128 × 128 test images

Table 2 Image quality metrics for Green plane of various color images

Embedding Scheme (ES) (1,n)	Girl				Peppers			
	P_E	P_{NE}	Payload	PSNR	P_E	P_{NE}	Payload	PSNR
1,1	2198	14,186	16,384	52.92	2119	14,265	16,384	52.88
1,2	2198	14,186	18,582	51.77	2119	14,265	18,503	51.90
1,3	2198	14,186	20,780	49.23	2119	14,265	20,622	49.35
1,4	2198	14,186	22,978	44.63	2119	14,265	22,741	44.63
1,5	2198	14,186	25,176	39.16	2119	14,265	24,860	38.60
1,6	2198	14,186	27,374	33.35	2119	14,265	26,979	33.13
Lena					Plane			
ES (1,n)	P_E	P_{NE}	Payload	PSNR	P_E	P_{NE}	Payload	PSNR
1,1	2268	14,116	16,384	52.83	2310	14,074	16,384	55.80
1,2	2268	14,116	18,652	51.96	2310	14,074	18,694	54.09
1,3	2268	14,116	20,920	49.34	2310	14,074	21,004	50.23
1,4	2268	14,116	23,188	44.56	2310	14,074	23,314	45.10
1,5	2268	14,116	25,456	39.10	2310	14,074	25,624	38.88
1,6	2268	14,116	27,724	32.2	2310	14,074	27,934	31.26

and Normalized Absolute Error (NAE). These indices are calculated by using the relations given below:

$$\mathrm{MSE} = \frac{1}{WH}\sum_{j=1}^{W}\sum_{k=1}^{H}\left(x_{j,k} - x'_{j,k}\right)^2 \tag{8}$$

Where W and H are Width and Height of cover image, $x_{j,k}$ and $x'_{j,k}$ are the pixel values of the cover and stego-images respectively.

$$\mathrm{PSNR} = 10\log\frac{(2^n-1)^2}{MSE} = 10\log\frac{(255)^2}{MSE} \tag{9}$$

Table 3 Image quality metrics for Blue plane of various color image

Embedding Scheme (ES) (1,n)	Girl				Peppers			
	PE	PNE	Payload	PSNR	PE	PNE	Payload	PSNR
1,1	2247	14,137	16,384	55.89	2247	14,137	16,384	55.74
1,2	2247	14,137	18,631	53.86	2247	14,137	18,631	54.23
1,3	2247	14,137	20,878	50.26	2247	14,137	20,878	50.23
1,4	2247	14,137	23,125	44.82	2247	14,137	23,125	44.82
1,5	2247	14,137	25,372	39.09	2247	14,137	25,372	39.31
1,6	2247	14,137	27,619	33.23	2247	14,137	27,619	33.31
Lena					Plane			
ES (1,n)	PE	PNE	Payload	PSNR	PE	PNE	Payload	PSNR
1,1	2429	13,955	16,384	55.76	2318	14,066	16,384	54.62
1,2	2429	13,955	18,813	53.86	2318	14,066	18,702	53.87
1,3	2429	13,955	21,242	49.91	2318	14,066	21,020	50.16
1,4	2429	13,955	23,617	44.65	2318	14,066	23,338	44.7
1,5	2429	13,955	26,100	38.92	2318	14,066	25,656	38.82
1,6	2429	13,955	28,529	33.34	2318	14,066	27,974	31.25

Table 4 Image quality metrics for Red plane of various color images

Cover Image	Lena	Plane	Peppers	Girl
PSNR (dB)	46.98	46.80	46.87	46.36

$$NAE = \frac{\sum_{j=1}^{M}\sum_{k=1}^{N}\left|x_{j,k}-x'_{j,k}\right|}{\sum_{j=1}^{M}\sum_{k=1}^{N}\left|x_{j,k}\right|} \tag{10}$$

$$NCC = \frac{\sum_{j=1}^{M}\sum_{k=1}^{N}x_{j,k}\times x'_{j,k}}{\sum_{j=1}^{M}\sum_{k=1}^{N}{x_{j,k}}^{2}} \tag{11}$$

6.1 Perceptual quality analysis

Firstly, the proposed scheme has been tested for various 128 × 128 sized test images for a varying payload as shown in Fig. 7.

The perceptual quality results for the mentioned images obtained using proposed work for Green, Blue and Red planes are shown in Tables 2, 3 and 4 respectively, while as the results of NCC and NAE are shown in Table 5. The number of edge and non-edge pixels in a given cover image have been denoted by P_E and P_{NE} respectively. Here the number of secret bits to be embedded in G and B planes of the cover image depends upon the Embedding Scheme (ES) {(1, 1); (1, 2); (1, 3)….; (1, n)}. It is pertinent to mention that by embedding scheme (1, n), we mean one bit of data is embedded in non-edge pixel and 'n' bits in an edge pixel. Further total payload is sum of logo bits and secret data bits. As an example for embedding scheme of (1,1) total payload (16,384) for green plane would comprise of 512 logo bits and rest would be (16,384-512 = 15,872) secret data bits, while as other 512 logo bits would be contained by blue plane of the same image.

It is evident from Tables 2 and 3 that our technique is capable of providing high quality stego-images even for a fairly high payload. This is substantiated by the fact that for average payload of 1.70 bits per pixel for each plane (B and G) the average PSNR of our technique is about 32.5 dB. Table 4 shows the PSNR for Red plane with data embedded in two LSBs. Further, from Table 5, it is clear that the value of NCC tends to unity and NAE values are less than 0.01. This indicates that the quality of stego-image obtained is better. The image quality

Table 5 Results of NCC and NAE

Embedding scheme	Lena		Peppers		Plane		Girl	
	NCC	NAE	NCC	NAE	NCC	NAE	NCC	NAE
1,1	0.9995	0.0015	0.9994	0.0018	0.9995	0.0011	0.9993	0.0033
1,2	0.9996	0.0017	0.9994	0.0020	0.9995	0.0013	0.9991	0.0043
1,3	0.9995	0.0020	0.9994	0.0024	0.9995	0.0014	0.9991	0.0049
1,4	0.9996	0. 0026	0.9994	0.0032	0.9995	0.0018	0.9991	0.0050
1,5	0.9996	0. 0038	0.9994	0.0047	0.9995	0.0020	0.9989	0.0065
1,6	0.9996	0. 0061	0.9994	0.0070	0.9995	0.0039	0.9987	0.0084

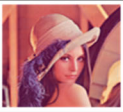
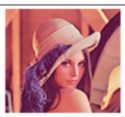
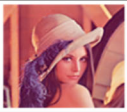
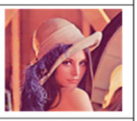
Es	1,1	1,2	1,3	1,4	1,5	1,6
Lena (stego)						
PSNR (dB)	52.52	51.47	50.33	46.84	41.88	36.07
Pepper (Stego)						
PSNR (dB)	52.57	52.00	50.34	46.67	41.67	35.61
Girl (Stego)						
PSNR (dB)	52.42	51.85	50.48	48.78	44.58	39.40
Plane (Stego)						
PSNR (dB)	52.45	51.85	50.22	46.65	41.54	35.84

Fig. 8 Quality of stego-image for different Embedding Schemes (ES)

(both subjective & objective) for various test images with a variable number of data bits embedded in edge pixels is shown in Fig. 8. From Fig. 8, it is clear that the quality of stego-images remain good even when 6 LSBs of edge pixel of the cover images are replaced by the secret message.

The proposed technique has been compared with that of [6] as shown in Table 6. This comparison is shown graphically in Fig. 9. From the comparative results it is clear that our method shows better results of PSNR than [6]. The average increase in PSNR is around 2.75 dB which means the quality of stego-image in the proposed system is better. This increase is due to the fact that, in Chen et al., method as number of pixels in a block increases the perceptual quality decreases because of more bits of status indicator pixel P1 are used to hold the status of n-1 pixels. However, in our technique number of bits required to hold the status of pixels (edged or non-edged) remains same.

Further, the results of the proposed method have been also compared with [2]. The comparison of results is shown in Table 7. For comparison purpose, one bit has been

Table 6 Comparison between the proposed method and Chen et al. [6]

Capacity in Bits	PSNR in dB [6]	PSNR in dB (Proposed)
8192	51.00	52.43
9427	50.00	51.87
10,662	47.1	50.42
11,897	42.3	46.92
13,132	36.9	41.92
143,67	31.3	36.12

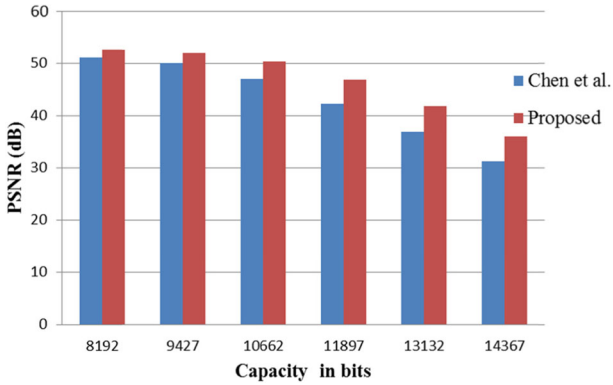


Fig. 9 Comparison of PSNR

embedded in non-edge pixel and three bits in edge pixel. The graphical representation of comparative results is shown in Fig. 10.

The results are compared for different values of payload. It is observed that the PSNR of the proposed method is better than [2]. The average increase in PSNR is 1.65 dB which indicates that the quality of stego-images in proposed method is better than that provided by scheme under comparison. The efficacy of the proposed technique has been also tested by histogram analysis. Figure 11 shows the histogram of constituent planes of cover image Lena, Peppers and Girl and their stego-images. The histograms of green and blue planes of the mentioned images have been presented for two different embedding techniques viz. {(1, 3) and (1, 6)}.

It is evident from Fig. 11 that an adversary cannot get a clue about data embedded from the analysis even if worst case embedding scheme (1, 6) is used. This feature makes the stego- images obtained using proposed method resistant to statistical attacks.

In addition to above tests the proposed scheme has been tested for ten randomly selected color images from UCID repository of size 384 × 512. Figure 12 shows the test and corresponding stego-images obtained using (1, 6) embedding technique. It is evident from the subjective quality of the stego images obtained that the proposed scheme is capable of providing high quality stego-images. Table 8 shows the objective quality indices obtained for (1, 6) embedding technique. The average values of PSNR, NCC and NAE testify the above argument.

One of the most commonly used image sizes for testing an image processing algorithm is 512 × 512. We have tested our algorithm for different 512 × 512 test images. The results obtained have been compared with few conventional already existing techniques. These results are shown in Table 9. It is in place to mention that LSB-3 is the conventional LSB scheme, wherein three bits of information are added in three LSBs of every pixel. It is evident from the averages values that our technique performs better.

Table 7 Comparison between the proposed method and Anastasia et al. [2]

Capacity in Bits	PSNR in dB [2]	PSNR in dB (proposed)
15,213	45.91	48.86
20,490	46.88	47.61
30,811	44.45	45.70

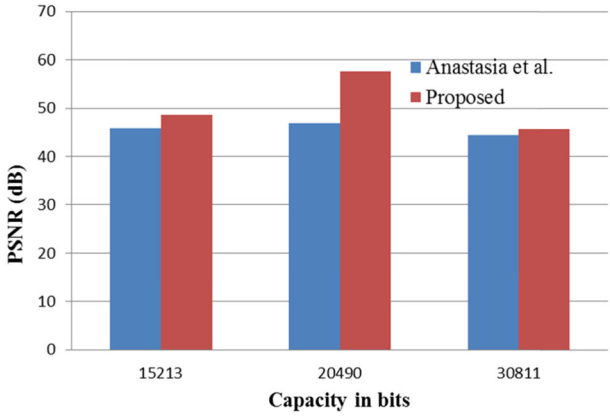


Fig. 10 Graphical results of PSNR

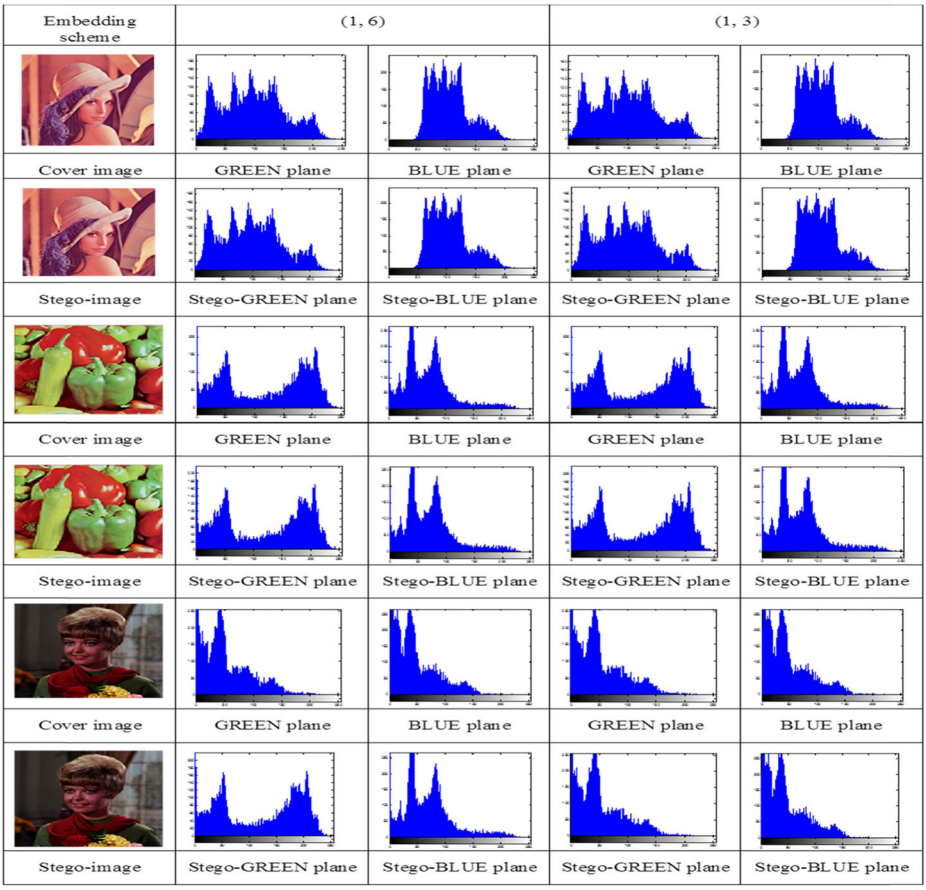


Fig. 11 Histograms of constituent planes of various cover images

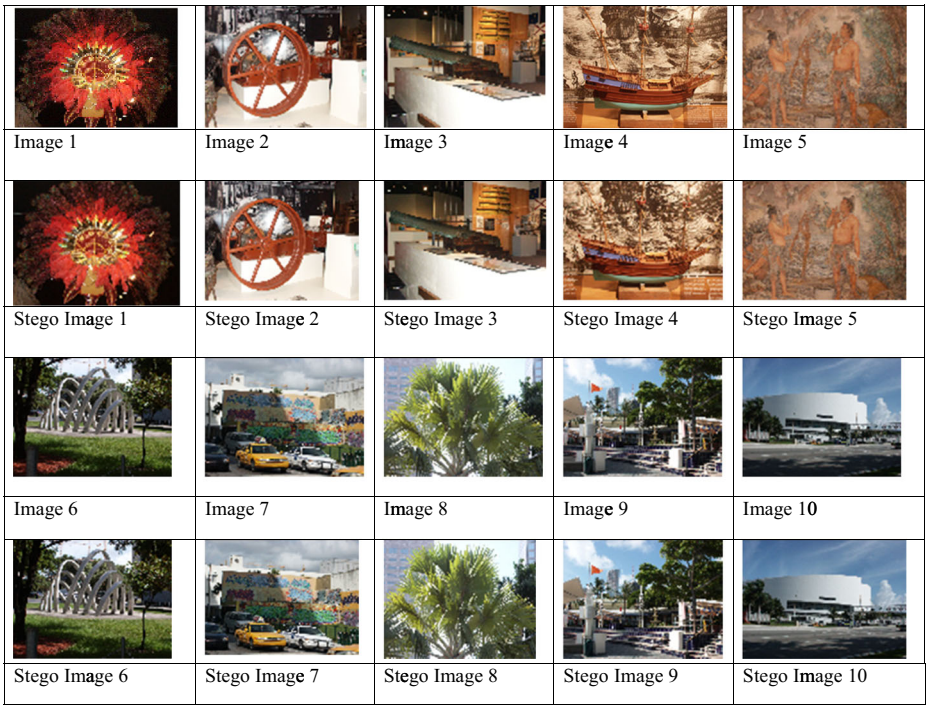


Fig. 12 Various UCID images and their respective Stego images for embedding scheme (1, 6)

6.2 Robustness analysis

In the proposed scheme the information hiding has been carried out in spatial domain. The stego- images obtained in our scheme have been tested to various image processing and geometric attacks like JPEG compression (which is one of the mostly used CODEC attacks), Additive white Gaussian Noise (AWGN), Salt and Pepper noise, Sharpening and Rotation. The results obtained for embedding scheme (1, 1) in terms of Bit Error Rate (BER) and NCC are shown in Table 10.

It is evident from Table 10 that even for very high Quality Factor (QF =90) value of JPEG compression (i.e. for a smaller degree of compression, as QF inversely varies with compression ratio) we are not able to retrieve the embedded logo correctly, indicating that our scheme is capable of detecting even a small degree of tamper. It is quite evident that decrease in QF would add more compression and hence more degradation of stego-image. This results in more deterioration of the subjective quality of the extracted logo and hence easier for the receiver to detect the tamper. Similar effects have been seen during experimentation by increasing the density of salt and pepper noise, variance of Additive White Gaussian Noise (AWGN) or increasing angle of rotation of the stego-image. Figure 13 shows various extracted logos from stego-image Lena for various attacks on it. Table 11 presents the PSNR computed between original and attacked test image “Lena” for embedding scheme of (1, 1) and that

Table 8 Image quality indices for various UCID images

Image	P_E	P_{NE}	Capacity (bits)	PSNR (dB)	NCC	NAE
Image 1	20,999	175,609	301,603	34.1166	0.9994	0.0079
Image 2	23,854	172,754	315,878	30.0425	0.9995	0.0029
Image 3	22,530	174,078	309,258	34.0783	0.9995	0.0031
Image 4	40,181	156,427	397,513	31.9457	0.9995	0.0037
Image 5	45,326	151,282	423,238	31.6689	0.9994	0.0038
Image 6	29,942	166,666	346,318	33.5126	0.9995	0.0041
Image 7	31,725	164,887	355,233	33.4545	0.9993	0.0027
Image 8	36,413	160,195	378,673	32.4233	0.9991	0.0023
Image 9	30,798	165,810	350,598	33.4755	0.9990	0.0032
Image10	20,622	175,986	299,718	35.0933	0.9993	0.0031

between embedded watermark and extracted one after various attacks. Very mall values of PSNR (between embedded and extracted watermarks) and highly deteriorated subjective quality of extracted watermarks, substantiates the fragility and tamper detection ability of our scheme.

6.3 Early tamper detection and content authentication

As shown from robustness analysis that proposed hiding scheme is fragile to various attacks. We have successfully put this adversity to use by embedding a logo besides secret data for authentication of the content at the receiver. When the logo embedded is received in its true form at the receiver it indicates that the secret data has not been fiddled with during transit from transmitter to receiver and vice versa. Further we have used the fragile nature of our scheme for early tamper detection as well, as discussed below:

One of the important applications of the information hiding schemes is covert communication. In certain critical applications pertaining to national security, the information extraction time is vital as real time decision making in such a scenario is linked with received information. Usually appreciable time is lost if receiver comes to know that extracted data has been compromised after extracting whole data from cover image. We have addressed this issue successfully, as our scheme is capable of informing about any possible attack on secret data during transit by only extracting a very small chunk of data, thus saving critical time of the receiver. In the proposed scheme we use a fragile watermark (logo) of size (32 × 32 bits) whose size is very small compared to secret data. At the receiver, firstly logo is extracted, if it is same as that embedded, then only secret data is extracted. On the other hand (in case logo is not received as embedded), the receiver asks for retransmission of stego-image containing secret data. To appreciate the amount of time which can be saved using this approach, Table 11

Table 9 Comparison with conventional techniques

Cover Image	LSB-3		Yu et al., method [11]		Proposed method	
	Capacity (bits)	PSNR (dB)	Capacity (bits)	PSNR (dB)	Capacity (bits)	PSNR (dB)
Lena	36,985	51.2655	37,348	55.5273	38,427	57.8723
Peppers	33,205	51.6637	34,263	56.3618	35,946	58.0023
Baboon	96,340	47.0073	102,574	51.0359	108,074	53.4945
Girl	96,340	46.9821	Not Available	Not Available	108,074	53.1230

Table 10 Robustness analysis to various attacks

Image → / Attacks ↓	Lena		Pepper		Girl		Plane	
	BER	NCC	BER	NCC	BER	NCC	BER	NCC
No Attack	0	1	0	1	0	1	0	1
JPEG QF = 90	0.4978	0.5107	0.5351	0.4505	0.5312	0.4638	0.4765	0.5252
AWGN Var = 0.0002	0.4746	0.5263	0.5097	0.50	0.5205	0.48	0.5117	0.4923
Salt and pepper Density = 0.01	0.0151	0.9090	0.065	0.9386	0.0693	0.9572	0.0761	0.9585
Sharpening (3x3)	0.50	0.4953	0.5175	0.4825	0.5107	0.4956	0.4501	0.5758
Rotation 5 degree	0.6162	0.3596	0.6328	0.3410	0.5898	0.3958	0.6201	0.3575

presents extraction times for logo and secret data from various 512 × 512 stego images (wherein (1, 1) embedding has been used). It is evident from the table that decision regarding tamper detection is on an average about eighteen times faster compared to conventional approach. The scheme of early tamper detection is more efficient when the ratio of secret data bits to logo bits is increased.

7 Conclusion

In this paper, a steganographic technique for color images using edge detection has been presented. The cover image (color) is partitioned into constituent R, G and B planes. Hybrid of Prewitt and Canny edge detection processes is used to find the edge pixels of G and B planes of cover image. The Red plane has been used to hold the status of G and B plane pixels. The data embedding has been carried out using a stego key to enhance security of embedded data. The proposed technique has been tested for three types of color images of sizes 128 × 128, 384 × 512 and 512× 512. It has been observed that besides providing two layers of security for embedded data (RC4 encryption and PAV controlled embedding) the proposed technique is capable of hiding a fair amount of payload while maintaining good imperceptibility. This is substantiated by high PSNR, near unity NCC and very small NAE values. Further, the scheme is capable of content authentication and early tamper detection. These features have been achieved by embedding a fragile watermark whose size is very small compared to size of

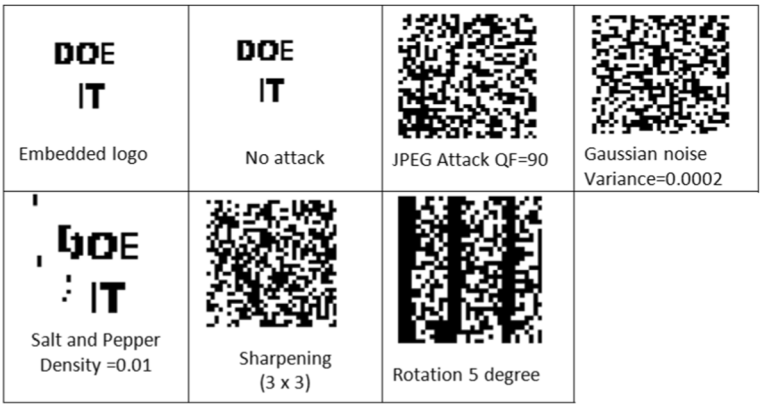


Fig. 13 Extracted logos from stego-image Lena after various attacks

Table 11 Comparison of secret data and logo extraction times

Image	Data extraction time (in seconds)	Logo extraction time (in seconds)
Lena	1.0000 sec.	0.0313 sec.
Plane	0.9375 sec.	0.0469 sec.
Peppers	0.9844 sec.	0.0781 sec.
Girl	1.0000 sec.	0.0625 sec.

secret data. The proposed technique has been compared with various state-of- art techniques and the results show that our technique performs better.

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